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Assignment 202

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IARCO RESEARCH PROPOSAL

Evaluating Early Cancer Detection and Tissue-of-Origin Classification Using cfRNA-Based Liquid Biopsy

Seeun PARK

Raffles American School (Grade 12)

Junior Category

Malaysia

October 5, 2026

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Title

Evaluating Early Cancer Detection and Tissue-of-Origin Classification Using cfRNA-Based Liquid Biopsy

Research Problem

Cancer is a leading cause of the world population's death in an era of aging global population, posing a major challenge to global public health: lung, breast, and colorectal cancers are the top 3 most common types [1]. Although risk factors causing cancer – tobacco use, alcohol consumption, and the lack of fresh produce – are well recognized, the truth is that the number of individuals suffering from cancer has not markedly declined. Official statistics project that the number of new cancer cases per year is expected to rise to 33 million and the number of cancer-related deaths to 18.2 million by 2050 [2].

Despite decades of effort, involving recent advances in prevention such as mRNA cancer vaccines, the detection of cancer is happening via various lab and imaging tests; however, they are not fully definitive. While tissue biopsy remains the sole definitive diagnostic method, which is the sole definitive indicator [3], an easier manageable (less invasive) and more broadly scalable method is necessary for accelerating the start of the treatment process. With prevention strategies after early diagnosis, we can lessen the burden of cancer on approximately 38% of patients [1].

In this proposal, we will present a blood-based liquid biopsy leveraging the unique properties of cfRNA (cell-free RNA) to enable earlier cancer detection and accurate identification of the cancer's origin tissue. By instituting a new diagnostic method utilizing blood-based cfRNA, rather than a traditional invasive test (tissue biopsy), we can ultimately enhance the patient survival outcomes of patients through early intervention.

Existing Literature

Recent studies suggest cfRNA as a potential biomarker for cancer testing. Cell-free RNA (cfRNA), the molecules that are present in bodily fluids – such as blood, urine, or cerebrospinal fluid, outside of cells, comprises a diverse array of RNA molecules – including mRNA, non-coding RNA, and fragmented RNA – resulting by cellular processes [4]. Industry report highlights its key advantages: tracing the tissue-of-origin, early screening for low tumor shedding, gene fusion detection, and even further monitoring treatment response through repeatable sampling [4].

Liquid biopsy captures several tumor-derived components, collectively termed tumor biomarkers – including CTC (circulating tumor cells), ctRNA/DNA (circulating tumor RNA/DNA) shed by dying tumor cells, DNA methylation, and cfRNA/DNA [5]. Among these, CTC analysis has been enabled by the application of genomics, transcriptomics, proteomics, and epigenetic profiling, providing a more comprehensive view of their molecular characteristics [6]. However, CTCs are rare and present in very low numbers in the bloodstream, especially in the

early stages, which limits sensitivity [6]. Similarly, recent work has demonstrated the utility of ctRNA/DNA profiling; gene fusions are inherently difficult to detect by DNA-only sequencing [7]. Integrating RNA (ctRNA and cfRNA) with ctDNA overcomes the barrier: combined analysis increases total actionable diagnostic yield by 36.7% and broadened panel coverage [7]. On the other hand, Jamshidi *et al*. [8] displayed the performance of DNA methylation profiling, which estimates the relative contribution of different tissues toward the mixture of cfDNA in biofluids, was the most sensitive cfRNA (compared with genotype, fragmentation profiles, and chromosome copy numbers). Nonetheless, DNA methylation analysis has important limitations. Conor Loy, *et al*. later mentioned the lack of a gold standard and limited reference methylomes, which hinder the reliable evaluation and accurate deconvolution, regardless of its great cancer origin identification [8].

To address these analytical limitations, Y. Tao, *et al.* reported that cfRNA profiling – a high-throughput sequencing method in which cfRNA is converted into cDNA (complementary DNA) for in-depth analysis [4] – demonstrates superior clinical sensitivity compared to cfDNA with respect to cancer detection [9]. Their study even disclosed outstanding signatures from cfRNA-seq data to specify cancer stage, tumor size, and survival. These findings suggest that cfRNA-based approaches may prove more sensitive and adaptable in clinical practice.

Research Questions

How effectively can a cfRNA-based liquid biopsy detect early-stage cancer and predict tissue-of-origin, compared to the conventional diagnostic approaches?

- Can cfRNA signatures reliably differentiate cancer patients from healthy controls?
- Can the model accurately pinpoint the exact organ of origin across 5 cancer types: colorectal, stomach, liver, lung, and esophageal cancer?
- Does the cfRNA biomarker panel maintain high sensitivity in early-stage (Stage I/II) detection?

Methodology

This study will adapt publicly available cfRNA sequencing data from NCBI GEO (GSE174302) to develop and evaluate a machine-learning-based liquid biopsy diagnosis framework, for early detection of cancer and cancer-type classification. The cancer-status and cancer-type annotations provided in the GEO metadata will be used as the primary reference labels; pathology-confirmed or biopsy-related information will be used as an additional reference, where available. The analysis will progress through standard transcriptomics and machine-learning AI: 1. Data acquisition and curation 2. Preprocessing and normalization 3. Feature selection 4. AI model training and validation 5. Performance evaluation against reference labels

1. Data acquisition and curation

- Download cfRNA sequencing data and clinical metadata
- Check the format and processing status of cfRNA expression data
- Check sample information and arrange into a chart
- Mark samples with missing or unclear information as "unknown" and will be excluded from a specific analysis when the missing or unclear information is necessary for the analysis
- Group them into cohorts:

13 Cancer vs non-cancer (control)

Early stage of cancer vs non-cancer (control)

Cancer-type groups – colorectal, stomach, liver, lung, and esophageal

2. Preprocessing and normalization

- Remove samples with missing, duplicated, poor quality values
- Remove lowly expressed genes across most samples
- Correct for differences in sequencing depth and library size between samples
- Normalize the data using the TMM method in the edgeR package to enable comparison across most samples (if the data is raw count)

3. Feature selection

- Divide the sample data into two sets, before model feature selection: training and testing
- Remove genes with little or no expression across using the training set only to prevent data leakage
- 7 Select important genes associated with cancer detection and cancer-type classification

7 Cancer detection: genes showing differential expression between cancer and non-cancer control samples will be selected

7 Early cancer detection: genes showing differential expression between early-stage cancer and non-cancer control samples will be selected

Cancer-type (tissue-of-origin) classification: genes showing distinct expression patterns across different cancer types will be selected

4. AI model training and validation

- 17 Candidate machine-learning algorithms, including Logistic Regression and Random Forest
- Train the AI using the training set

- Verify the model stability through cross-validation within the training set
- Evaluate the final model using the independent testing set

5. Performance evaluation against reference labels

- Select only samples with available biopsy records; exclude samples lacking such confirmation because the ground truth cannot be defined.
- Performance detection:

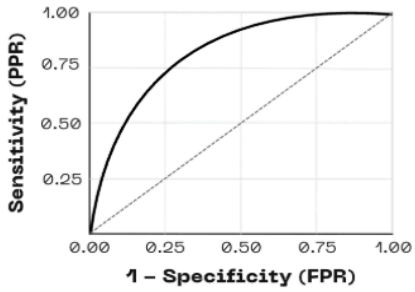
  Sensitivity & Specificity: the proportions of true positive cancer cases and true negative control cases correctly identified

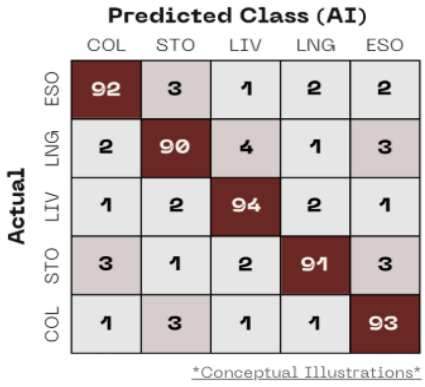
  Accuracy & F1-score: overall classification accuracy and harmonic mean of precision/recall

  AUC (ROC Curve): overall discriminative ability between cancer and non-cancer states

  Confusion matrix: a multi-class classification breakdown showing tissue-of-origin prediction performance across the 5 cancer types

Expected Values





We anticipate that the cfRNA-based machine learning framework will achieve high clinical sensitivity and specificity for early cancer detection. Specifically, for multi-class tissue-of-origin classification across colorectal, stomach, liver, lung, and esophageal cancers, the model is targeted to achieve AUC > 0.90, demonstrating high discrimination between organ-specific cancer signals.

Research Topic

Cancer is often detected only after it has progressed to an advanced stage, whereas early detection can improve treatment response and survival outcomes. To catch oncogenic signals

before disease progression, a non-invasive and accessible approach is suggested as a feasible solution. This research focuses on detecting cancer at an early stage and tissue-of-origin classification utilizing cfRNA liquid biopsy. By integrating public cfRNA-seq datasets (GSE174302) with bioinformatics feature selection and machine-learning algorithms, this study explores the potential of cfRNA expression patterns as computational biomarkers for cancer diagnosis.

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Quality of Writing

The language of this proposal is academically precise, clear, and appropriate for multidisciplinary audiences. This proposal integrates biological concepts with computational methods, making the proposal comprehensible to readers from both biology and engineering fields. The use of the publicly available dataset GSE174302, open-source tools, and evaluation metrics supports the feasibility of the methodology. The proposal also follows a standard research structure and adheres to IARCO's requirements – grammar, formatting, and IEEE citation style.

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